



Solution of Triangles: RACE-01

Single Correct Question

[Note: All symbols used have usual meaning in triangle.]

1. The real value of x such that $x^2 + 2x$, $2x + 3$ and $x^2 + 3x + 8$ are lengths of the sides of a triangle is equal to
 (A) $(5, \infty)$ (B) $(4, \infty)$ (C) $(3, \infty)$ (D) $(0, \infty)$ MTS117
2. In a $\triangle ABC$, if $\angle A = \frac{\pi}{4}$, $\angle B = \frac{5\pi}{12}$, then $\frac{a+c\sqrt{2}}{b}$ is equal to
 (A) 1 (B) 2 (C) 3 (D) 4 MTS118
3. If in a $\triangle ABC$, $\angle A = \frac{\pi}{6}$ and $b : c = 2 : \sqrt{3}$, then $\angle B =$
 (A) 30° (B) 45° (C) 90° (D) 60° MTS119
4. In any $\triangle ABC$, $a \sin(B - C) + b \sin(C - A) + c \sin(A - B) =$
 (A) 1 (B) 0 (C) 3 (D) 5 MTS120
5. The perimeter of a triangle ABC is six times the arithmetic mean of the sines of its angles. If the side b is equal to $\sqrt{3}$, then angle $B =$
 (A) 30° (B) 45° (C) 90° (D) 60° MTS121
6. If $a : b : c = 4 : 5 : 6$, and $\angle C = \lambda \angle A$, then $\lambda =$
 (A) 1 (B) 2 (C) 3 (D) 4 MTS122
7. In a $\triangle ABC$, if $9a^2 + 9b^2 = 19c^2$, then the value of $\frac{\cot C}{\cot A + \cot B}$ is equal to
 (A) $\frac{1}{9}$ (B) $\frac{2}{9}$ (C) $\frac{5}{9}$ (D) $\frac{7}{9}$ MTS123
8. In a triangle ABC , the sides a, b, c are roots of $x^3 - 11x^2 + 38x - 40 = 0$. If $\sum \left(\frac{\cos A}{a} \right) = \frac{p}{q}$, then the least value of $(p + q)$ where $p, q \in N$ is equal to
 (A) 10 (B) 15 (C) 20 (D) 25 MTS124

9. In a triangle ABC , $a : b : c = 4 : 5 : 6$. Then $3A + B$ equals to
 (A) $4C$ (B) 2π (C) $\pi - C$ (D) π

MTS125

10. If in a triangle ABC , $\sin A, \sin B$ & $\sin C$ are in A.P. ($C > 90^\circ$), then $\cos A$ is
 (A) $\frac{3c-4b}{2b}$ (B) $\frac{3c-4b}{2c}$ (C) $\frac{4c-3b}{2b}$ (D) $\frac{4c-3b}{2c}$

MTS126

Subjective Type Question

11. In any triangle ABC , if $\frac{a^2-b^2}{a^2+b^2} = \frac{\sin(A-B)}{\sin(A+B)}$, then prove that the triangle is either right angled or isosceles.

MTS127

12. If the angles of $\triangle ABC$ are in the ratio $1 : 2 : 3$, then find the ratio of their corresponding sides

MTS128

13. In any $\triangle ABC$, prove that $\frac{b-c}{b+c} = \frac{\tan\left(\frac{B-C}{2}\right)}{\tan\left(\frac{B+C}{2}\right)}$.

MTS129

14. If in a triangle ABC , $\angle C = 60^\circ$, then prove that $\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$.

MTS130

15. In a $\triangle ABC$ prove that $\frac{c}{a-b} = \frac{\tan\frac{A}{2} + \tan\frac{B}{2}}{\tan\frac{A}{2} - \tan\frac{B}{2}}$

MTS131

16. In $\triangle ABC$, if A, B, C are in A.P., then prove that $2\cos\frac{A-C}{2} = \frac{a+c}{\sqrt{a^2+c^2-ac}}$.

MTS132

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	C	B	D	B	C	D	D	D

12. $a : b : c = \sin A : \sin B : \sin C = 1 : \sqrt{3} : 2$



Solution of Triangles: RACE-02

Single Correct Question

[Note: All symbols used have usual meaning in triangle.]

1. In $\triangle ABC$, the value of $\frac{\cos A + \cos C}{a+c} + \frac{\cos B}{b} =$
 (A) $\frac{1}{b}$ (B) $\frac{2}{b}$ (C) $\frac{3}{b}$ (D) $\frac{4}{b}$ MTS133
2. In $\triangle ABC$, the value of $a(b^2 + c^2)\cos A + b(c^2 + a^2)\cos B + c(a^2 + b^2)\cos C =$
 (A) abc (B) $2abc$
 (C) $3abc$ (D) $4abc$ MTS134
3. In $\triangle ABC$, the value of $(b+c)\cos A + (c+a)\cos B + (a+b)\cos C =$
 (A) $a-b+c$ (B) $a+b-c$
 (C) $a-b-c$ (D) $a+b+c$ MTS135
4. If in a $\triangle ABC$, $b = 3c$, then the value of $\cot \frac{A}{2} \cdot \cot \frac{B-C}{2}$ is equal to
 (A) 1 (B) 2 (C) 4 (D) 6 MTS136
5. In a triangle ABC , $\angle A = 60^\circ$ and $b:c = \sqrt{3}+1:2$, then the value of $(\angle B - \angle C)$ is equal to
 (A) 15° (B) 30° (C) 45° (D) 60° MTS137
6. In any $\triangle ABC$, $(abc) \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} =$
 (A) Δ^2 (B) Δ (C) Δ^3 (D) $\frac{1}{\Delta}$ MTS138
7. If in a triangle ABC , $\Delta = a^2 - (b-c)^2$, then the value of $\tan A =$
 (A) $\frac{2}{15}$ (B) $\frac{4}{15}$ (C) $\frac{8}{15}$ (D) $\frac{7}{15}$ MTS139
8. In a triangle ABC , if $\tan \frac{A}{2} \cdot \tan \frac{B}{2} = \frac{1}{3}$ and $ab = 4$, then the least value of c is equal to
 (A) 1 (B) 2 (C) 3 (D) 5 MTS140

9. In ΔABC , if a, b, c (taken in that order) are in A.P. then the value of $\cot \frac{A}{2} \cot \frac{C}{2}$ is equal to
 (A) 5 (B) 7 (C) 3 (D) 9

MTS141

Subjective Type Question

10. In a ΔABC , prove that $\frac{\cos A}{c \cos B + b \cos C} + \frac{\cos B}{a \cos C + c \cos A} + \frac{\cos C}{a \cos B + b \cos A} = \frac{a^2 + b^2 + c^2}{2abc}$.

MTS142

11. ΔABC with $\angle A$ acute, $b = 2$ and $c = (\sqrt{3} - 1)$ has area $(\sqrt{3} - 1)/2$, then find the measures of angles of triangle.

MTS143

12. If A, B, C are the angle of a triangle, then prove that $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \frac{s^2}{\Delta}$

MTS144

13. In any triangle ABC , prove that $(a + b + c) \left(\tan \frac{A}{2} + \tan \frac{B}{2} \right) = 2c \cot \frac{C}{2}$.

MTS145

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	A	C	D	B	B	A	C	B	C

11. $\angle A = 30^\circ, \angle B = 135^\circ, \angle C = 15^\circ$



Solution of Triangles: RACE-03

Single Correct Question:

[Note: All symbols used have usual meaning in triangle.]

- In a $\triangle ABC$, the value of $r \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$ is equal to
 (A) $\frac{\Delta}{R}$ (B) $\frac{\Delta}{2R}$ (C) $\frac{\Delta}{3R}$ (D) $\frac{\Delta}{4R}$ MTS146
- If the median AD of a $\triangle ABC$ makes an angle $3\pi/4$ with the side BC , then evaluate $|\cot B - \cot C|$.
 (A) 4 (B) 2 (C) 1 (D) $\sqrt{3}$ MTS147
- If in a $\triangle ABC$, $AB = 4$ cm, $AC = 8$ cm and $\cos\left(\frac{B-C}{2}\right) = \frac{1}{\sqrt{2}}$; then evaluate $|2\cot B - \cot C|$.
 (A) 1 (B) 2 (C) 3 (D) 4 MTS148
- In $\triangle ABC$, if circumradius ' R ' and inradius ' r ' are connected by relation $R^2 - 4Rr + 8r^2 - 12r + 9 = 0$, then semi perimeter of the $\triangle ABC$ is equal to
 (A) $\frac{9\sqrt{3}}{2}$ (B) $\frac{5\sqrt{3}}{2}$ (C) $\frac{7\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2}$ MTS149
- In a $\triangle ABC$, prove that $a \cos B \cos C + b \cos C \cos A + c \cos A \cos B =$
 (A) $\frac{\Delta}{R}$ (B) $\frac{\Delta}{2R}$ (C) $\frac{2\Delta}{R}$ (D) $\frac{R}{\Delta}$ MTS150
- The ratio of the area of n -sided regular polygon, circumscribed about a circle, to the area of the regular polygon of equal number of sides inscribed in the circle is 4 : 3. The value of n is equal to
 (A) 3 (B) 6 (C) 9 (D) 10 MTS151
- A regular pentagon and a regular decagon have the same perimeter, then their areas are in the ratio.
 (A) $\frac{1}{\sqrt{5}}$ (B) $\frac{\sqrt{5}}{2}$ (C) $\frac{3}{\sqrt{5}}$ (D) $\frac{2}{\sqrt{5}}$ MTS152
- If the area of circle is A_1 and area of regular pentagon inscribed in the circle is A_2 , then $\frac{A_1}{A_2} =$
 (A) $\frac{9\pi}{5\sqrt{10+2\sqrt{5}}}$ (B) $\frac{7\pi}{5\sqrt{10+2\sqrt{5}}}$ (C) $\frac{8\pi}{5\sqrt{10+2\sqrt{5}}}$ (D) $\frac{4\pi}{5\sqrt{10+2\sqrt{5}}}$ MTS153

9. A circle is inscribed in a regular octagon whose area is Δ . If area of a new regular octagon formed by points of contact of polygon and incircle is Δ' , then $\frac{\Delta}{\Delta'}$ is

(A) $\cos^2 \frac{\pi}{8}$ (B) $\sec \frac{\pi}{8}$ (C) $\sec^2 \frac{\pi}{8}$ (D) $\cot^2 \frac{\pi}{8}$

MTS154

10. Let $A_i, i = 1, 2, 3, \dots, 10$ are vertices of regular decagon P . If area of polygon P is $5\sin\left(\frac{\pi}{5}\right)$ square units then the value of $(A_1A_2)(A_1A_3)(A_1A_4)\dots(A_1A_{10})$ is

(A) 10 (B) 9 (C) $45\sin\left(\frac{\pi}{5}\right)$ (D) $45\cos\left(\frac{\pi}{5}\right)$

MTS155

Subjective Type Question

11. Let Δ & Δ' denote the areas of a Δ and that of its incircle. Prove that

$$\Delta : \Delta' = \left(\cot \frac{A}{2} \cdot \cot \frac{B}{2} \cdot \cot \frac{C}{2} \right) : \pi$$

MTS156

12. In a ΔABC , prove that $\sin 2A + \sin 2B + \sin 2C = 2\Delta/R^2$

MTS157

13. In a ΔABC , prove that $\frac{b^2 - c^2}{2a} = R \sin(B - C)$

MTS158

14. In a ΔABC , show that $a + b + c = \frac{abc}{2Rr}$

MTS159

15. If the perimeter of a circle and a regular polygon of n sides are equal, then prove that

$$\frac{\text{area of the circle}}{\text{area of polygon}} = \frac{\tan \frac{\pi}{n}}{\frac{\pi}{n}}$$

MTS160

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	C	A	A	B	D	C	C	A